

The Sharp Queried-Gradient Constant of Nesterov's Fast Gradient Method at Horizon Two

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China

xiaojianzeng@foxmail.com

Abstract

For Nesterov's fast gradient method applied to convex differentiable functions with an L -Lipschitz gradient, consider the smallest gradient norm among the three queried points generated at horizon two. We determine its worst-case value under the initial condition $\|x_0 - x_*\| \leq R$ in every finite dimension. If

$$\beta = \frac{\sqrt{5} - 1}{1 + \sqrt{7 + 2\sqrt{5}}}, \quad c_2 = \frac{1}{3 + \beta},$$

then the squared worst-case value is exactly $c_2^2 L^2 R^2$. The upper bound is proved by an explicit smooth-convex interpolation certificate whose 4×4 Gram slack is positive semidefinite by exact algebraic inequalities. A one-dimensional projection-envelope function attains the bound. This supplies the analytic horizon-two branch left outside the previous horizon-one and horizon-at-least-seven results.

1 Problem and result

Performance estimation problems encode finite-horizon worst-case questions for first-order methods as semidefinite programs [1, 4]. For the queried-gradient criterion of Nesterov's fast gradient method (FGM), Du proved exactness of a known relaxed bound from horizon seven onward and treated horizon one separately; the five analytic branches $2 \leq N \leq 6$ were left uncharacterized [2]. We settle the first of those branches. The argument uses the complete smooth-convex interpolation inequalities, rather than identifying the relaxed bound of Kim and Fessler [3] with the exact problem.

At a methodological level, exact positive-semidefinite Gram certificates have also been used to establish sharply delimited constants in unrelated matrix and operator inequalities [6, 5]. Those works supply no premise below; the comparison concerns only the use of fully displayed algebraic certificates instead of floating-point evidence.

Let $\mathcal{F}_{0,L}(\mathbb{R}^d)$ be the class of convex differentiable functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ whose gradients are L -Lipschitz. Starting from $y_0 = x_0$ and $t_0 = 1$, FGM performs

$$\begin{aligned} y_{k+1} &= x_k - \frac{1}{L} \nabla f(x_k), \\ t_{k+1} &= \frac{1 + \sqrt{1 + 4t_k^2}}{2}, \\ x_{k+1} &= y_{k+1} + \frac{t_k - 1}{t_{k+1}}(y_{k+1} - y_k). \end{aligned}$$

At horizon two these updates are made for $k = 0, 1$, and gradients are queried at x_0, x_1, x_2 . Define

$$W_2(L, R, d) = \sup_{f \in \mathcal{F}_{0,L}(\mathbb{R}^d), \substack{x_* \in \arg \min f \\ \|x_0 - x_*\| \leq R}} \min_{0 \leq k \leq 2} \|\nabla f(x_k)\|^2.$$

Writing

$$t_1 = \frac{1 + \sqrt{5}}{2}, \quad t_2 = \frac{1 + \sqrt{7 + 2\sqrt{5}}}{2}, \quad \beta = \frac{t_1 - 1}{t_2} = \frac{\sqrt{5} - 1}{1 + \sqrt{7 + 2\sqrt{5}}},$$

we prove the following full-dimensional statement, including the degenerate radius boundary.

Theorem 1. *For every integer $d \geq 1$, every $L > 0$, and every $R \geq 0$,*

$$W_2(L, R, d) = c_2^2 L^2 R^2, \quad c_2 = \frac{1}{3 + \beta}.$$

In particular, $c_2^2 = 0.0928513193578614102140718294340 \dots$

The contribution relative to [2] is confined to $N = 2$. It does not address the post-gradient points y_k , the terminal gradient alone, or the four remaining horizons $3 \leq N \leq 6$.

2 An exact upper certificate

We first normalize $L = R = 1$, translate x_* to the origin, and subtract a constant so that $f_* = 0$. Set

$$a = 1 + \beta, \quad c = \frac{1}{a + 2} = c_2, \quad g_i = \nabla f(x_i), \quad g_* = 0.$$

The horizon-two recurrence becomes

$$x_1 = x_0 - g_0, \quad x_2 = x_0 - g_0 - ag_1. \quad (1)$$

For distinct i, j in $I = \{*, 0, 1, 2\}$, let

$$h_{ij} = f_i - f_j - \langle g_j, x_i - x_j \rangle - \frac{1}{2} \|g_i - g_j\|^2. \quad (2)$$

Every convex differentiable 1-smooth function satisfies $h_{ij} \geq 0$. Indeed, apply the unit-step descent inequality to $f - \langle g_j, \cdot \rangle$, which is minimized at x_j . These inequalities are also the standard exact interpolation conditions [4].

Let $\Gamma \succeq 0$ be the Gram matrix of the ordered vectors (x_0, g_0, g_1, g_2) . Denote their coordinate vectors by the standard basis of $\mathbb{R}^4$. The coordinates X_i, G_i of x_i, g_i implied by (1) are

i	$X_i^\top$					$G_i^\top$			
*	0	0	0	0		0	0	0	0
0	1	0	0	0		0	1	0	0
1	1	-1	0	0		0	0	1	0
2	1	-1	-a	0		0	0	0	1

where $\text{sym}(u, v) = (uv^\top + vu^\top)/2$. Consequently the Gram coefficient of (2) is

$$H_{ij} = -\text{sym}(G_j, X_i - X_j) - \frac{1}{2} \text{sym}(G_i - G_j, G_i - G_j), \quad h_{ij} = f_i - f_j + \langle H_{ij}, \Gamma \rangle_F.$$

Here $\langle A, B \rangle_F = \text{tr}(A^\top B)$.

Put

$$\tau = \min_{0 \leq k \leq 2} \|g_k\|^2, \quad r = 1 - \|x_0\|^2, \quad p_k = \|g_k\|^2 - \tau.$$

Thus $r, p_0, p_1, p_2 \geq 0$. All interpolation multipliers not displayed below are zero; in particular, $\lambda_{0*} = \lambda_{1*} = 0$:

ij	λ_{ij}	ij	λ_{ij}
01	$\frac{399a^2 - 201a + 2}{500(a+2)}$	02	$\frac{996 - 456a - 227a^2}{500(a+2)}$
10	$\frac{172a^2 - 485a + 342}{500(a+2)}$	12	$\frac{227a + 229}{500}$
20	$\frac{25}{144 - 53a}$	21	$\frac{250}{93}$
*0	$\frac{125(a+2)}{119}$	*1	$\frac{500}{2}$
2	$\frac{119}{500}$	2	$\frac{2}{a+2}$

(3)

The performance multipliers and radius multiplier are

$$(\mu_0, \mu_1, \mu_2) = \left(\frac{1}{1000}, \frac{1}{1000}, \frac{499}{500} \right), \quad \sum_{k=0}^2 \mu_k = 1, \quad \rho = c^2. \quad (4)$$

Let E_{rr} denote the 4×4 coordinate matrix unit, with coordinate 0 corresponding to x_0 and coordinates $k+1$ to g_k . Exact expansion of

$$S = \rho E_{00} - \sum_{k=0}^2 \mu_k E_{k+1, k+1} - \sum_{i \neq j} \lambda_{ij} H_{ij}$$

gives

$$S = \begin{pmatrix} \frac{1}{(a+2)^2} & \frac{53a-144}{250(a+2)} & -\frac{93}{1000} & -\frac{119}{1000} \\ \frac{53a-144}{250(a+2)} & \frac{46-17a}{40(a+2)} & \frac{-106a^2+249a-78}{500(a+2)} & \frac{79}{1000} \\ -\frac{93}{1000} & \frac{-106a^2+249a-78}{500(a+2)} & \frac{186a^2-829a+1598}{1000(a+2)} & \frac{119a^2+703a-1070}{1000(a+2)} \\ -\frac{119}{1000} & \frac{79}{1000} & \frac{119a^2+703a-1070}{1000(a+2)} & \frac{694-153a}{500(a+2)} \end{pmatrix}. \quad (5)$$

The function-value coefficients cancel in (3), and direct coefficient collection in the function values, Γ , and τ yields the certificate identity

$$c^2 - \tau = c^2 r + \sum_{k=0}^2 \mu_k p_k + \sum_{\substack{i,j \in I \\ i \neq j}} \lambda_{ij} h_{ij} + \langle S, \Gamma \rangle_F. \quad (6)$$

We next certify that every coefficient on its right-hand side is nonnegative.

Lemma 1. *All multipliers in (3)–(4) are positive and the matrix S in (5) is positive semidefinite.*

Proof. The specified radical a satisfies

$$a^4 - 8a^3 + 10a^2 - a - 1 = 0, \quad \frac{1281}{1000} < a < \frac{641}{500}. \quad (7)$$

For example, the enclosure follows by rational squaring from

$$\frac{559}{250} < \sqrt{5} < \frac{2237}{1000}, \quad \frac{3387}{1000} < \sqrt{7 + 2\sqrt{5}} < \frac{847}{250}.$$

It gives $281/1000 < \beta < 282/1000$. The only multiplier numerators requiring more than immediate positivity obey

$$\begin{aligned} 172(641/500)^2 - 485(641/500) + 342 &= \frac{91129}{31250} > 0, \\ 996 - 456(641/500) - 227(641/500)^2 &= \frac{9582013}{250000} > 0, \\ 144 - 53(641/500) &= \frac{38027}{500} > 0. \end{aligned}$$

Monotonicity on the interval in (7) justifies the indicated endpoints; the remaining numerators are plainly positive there.

For the matrix claim, direct reduction modulo (7) gives

$$S(1, c, c, c)^\top = 0. \quad (8)$$

Let $B = S[1:4, 1:4]$ be the lower-right 3×3 principal block. Its leading principal minors are

$$\Delta_1 = \frac{46 - 17a}{40(a + 2)}, \quad \Delta_2 = \frac{R_2(a)}{10^6(a + 2)^2}, \quad \Delta_3 = \frac{R_3(a)}{2 \cdot 10^8(a + 2)^2},$$

where

$$\begin{aligned} R_2(a) &= -227450a^3 + 701517a^2 - 1522068a + 1768420, \\ R_3(a) &= 4682448a^3 + 24778142a^2 - 115934851a + 124473901. \end{aligned}$$

Writing $\ell = 1281/1000$ and $u = 641/500$, termwise outward bounds give

$$\begin{aligned} R_2(a) &> -227450u^3 + 701517\ell^2 - 1522068u + 1768420 = \frac{2445271097527}{5000000} > 0, \\ R_3(a) &> 4682448\ell^3 + 24778142\ell^2 - 115934851u + 124473901 = \frac{205845508080081}{7812500} > 0. \end{aligned}$$

Also $\Delta_1 > 0$, so Sylvester's criterion gives $B \succ 0$. If $\mathbf{1} = (1, 1, 1)^\top$, the kernel identity (8) implies for every scalar u_0 and $z \in \mathbb{R}^3$ that

$$(u_0, z)^\top S(u_0, z) = (z - cu_0\mathbf{1})^\top B(z - cu_0\mathbf{1}) \geq 0.$$

Thus $S \succeq 0$ (and in fact has rank three). □

Because $S, \Gamma \succeq 0$, their Frobenius inner product is nonnegative. Every other term on the right of (6) is nonnegative by construction and Lemma 1. Hence $\tau \leq c^2$. We have proved

$$W_2(1, 1, d) \leq c^2 \quad \text{for every finite } d \geq 1. \quad (9)$$

The certificate depends only on a Gram matrix and therefore introduces no rank or dimension restriction.

3 A matching projection-envelope instance

Fix any unit vector $e \in \mathbb{R}^d$ and set $K = [0, ce]$. Define

$$f(x) = \max_{g \in K} \left\{ \langle x, g \rangle - \frac{1}{2} \|g\|^2 \right\}. \quad (10)$$

The maximizer is the Euclidean projection of x onto K , so $\nabla f(x) = \text{Proj}_K(x)$. Projection is firmly nonexpansive; consequently f is convex and 1-smooth. Moreover, $x_* = 0$ is a minimizer.

Start from $x_0 = e$. Since $c < 1/2$, the first queried gradient is $g_0 = ce$, and

$$x_1 = (1 - c)e, \quad g_1 = ce.$$

Using (1) and $(a + 2)c = 1$ gives

$$x_2 = (1 - c - ac)e = (1 - (a + 1)c)e = ce, \quad g_2 = ce.$$

Thus all three queried gradients have norm c , proving $W_2(1, 1, d) \geq c^2$. Together with (9), this establishes the normalized equality in every dimension.

For $R > 0$, translate the minimizer and apply the scaling

$$f_{L,R}(x) = LR^2 f\left(\frac{x - x_*}{R}\right).$$

Its gradient is L -Lipschitz, the FGM trajectory is the translated R -scaled normalized trajectory, and the squared gradients acquire the factor $L^2 R^2$. Conversely, any feasible instance with $R > 0$ is reduced to the normalized one by $\tilde{f}(z) = f(x_* + Rz)/(LR^2)$; its gradients and iterates scale by $1/(LR)$ and $1/R$, respectively. Thus the upper bound scales by the same factor. If $R = 0$, then $x_0 = x_*$, differentiable convex optimality gives $\nabla f(x_*) = 0$, and every subsequent iterate remains at x_* . Both sides of Theorem 1 are then zero. This completes the proof.

Reproducibility and assistance disclosure

An accompanying exact-arithmetic program reconstructs all twelve ordered interpolation matrices from (2), verifies function-value and Gram coefficient cancellation in (6), checks (7)–(8) and the three Sylvester minors, and recomputes the lower-witness recurrence. It can be run from the source distribution with

`python3 evidence/verify_exact_dual.py.`

The program uses symbolic expressions and rational comparisons; it is an audit of the displayed proof rather than a numerical premise. Floating-point semidefinite optimization was used only to discover a candidate sparse dual support.

A large-language-model coding assistant was used during exploratory dual search, symbolic-checking code development, and preparation of the exposition. The author checked the resulting mathematical statements and takes responsibility for the article. No theorem here is inferred from a model output, a numerical optimum, or a floating-point duality gap.

References

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